

AI-Based Fake News Detection System

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Introduction

- ✓ **Fake News Definition:** Refers to false or fabricated information presented as verified, genuine news.
- ✓ **Rapid Propagation:** Social media and digital communication platforms allow rumors and unverified news to spread globally in seconds.
- ✓ **Unconscious Belief:** Readers often accept and share misleading articles without prior fact-checking.
- ✓ **AI & ML Solution:** Artificial Intelligence and NLP provide automated tools to evaluate textual credibility accurately.



Problem Statement



⚠️ The Growing Challenge

Enormous volumes of news articles and posts are generated online every minute. Manual verification of every headline by human fact-checkers is slow, expensive, and practically impossible at Internet scale.

✳️ Impact of Misinformation

Uncontrolled fake news spreads fear, political manipulation, economic loss, and public distrust in legitimate media institutions. An intelligent, automated detection system is essential to tackle this issue.

Key Project Objectives



- ✓ **Automated Detection:** Build an automated system to classify news articles as REAL or FAKE.
- ✓ **Text Analysis:** Perform rigorous Natural Language Processing (NLP) on raw textual news content.
- ✓ **ML Classification:** Leverage supervised Machine Learning algorithms for high-accuracy binary decision making.
- ✓ **User-Friendly UI:** Develop an intuitive web interface for effortless real-time testing and result display.

Proposed System Pipeline



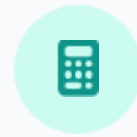
1. Input News

User inputs or pastes the news headline/article into the application.



2. Preprocessing

Cleans text, converts to lowercase, removes punctuation & stop words.



3. Feature Extraction

TF-IDF converts textual data into meaningful numeric vector matrices.



4. ML Prediction

Classifier model evaluates features and outputs REAL or FAKE label.

Technologies & Tools

Domain	Technology / Tool	Functionality & Purpose
Language	Python 3.x	Core programming language for ML pipeline and backend logic
NLP & ML	Scikit-Learn, NLTK	Text preprocessing, TF-IDF vectorization, Logistic Regression model
Web Interface	Streamlit / HTML5 / CSS3	Interactive, web-based dashboard for seamless user experience
Dataset	Labelled News Dataset	Curated dataset containing verified Real and Fake news articles

Working Methodology



STEP 01

Data Collection

Gathering labelled news dataset comprising thousands of authentic and fabricated news samples.

STEP 02

Preprocessing

Lowercasing, stripping special characters, stemming, and filtering out common stop words.

STEP 03

TF-IDF Vectorization

Converting textual tokens into TF-IDF term frequency numerical matrices.

STEP 04

Model & Inference

Training Logistic Regression classifier to evaluate new incoming article inputs in real-time.

Machine Learning Classification

95%

CLASSIFICATION
ACCURACY

Logistic Regression Classifier

Logistic Regression models the probability of binary outcomes (0 for Real, 1 for Fake) using the sigmoid activation function:

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(w^T X + b)}}$$

Offers high execution speed, low memory overhead, and proven efficiency for high-dimensional TF-IDF text feature spaces.

System Architecture



User Interface

Accepts raw text or web
links from the user



Text Preprocessing

Tokenization & TF-IDF
Feature Extraction



Trained Classifier

Logistic Regression pattern
matching



Output Result

Classification (REAL ● /
FAKE ●)

User Interface & Dashboard

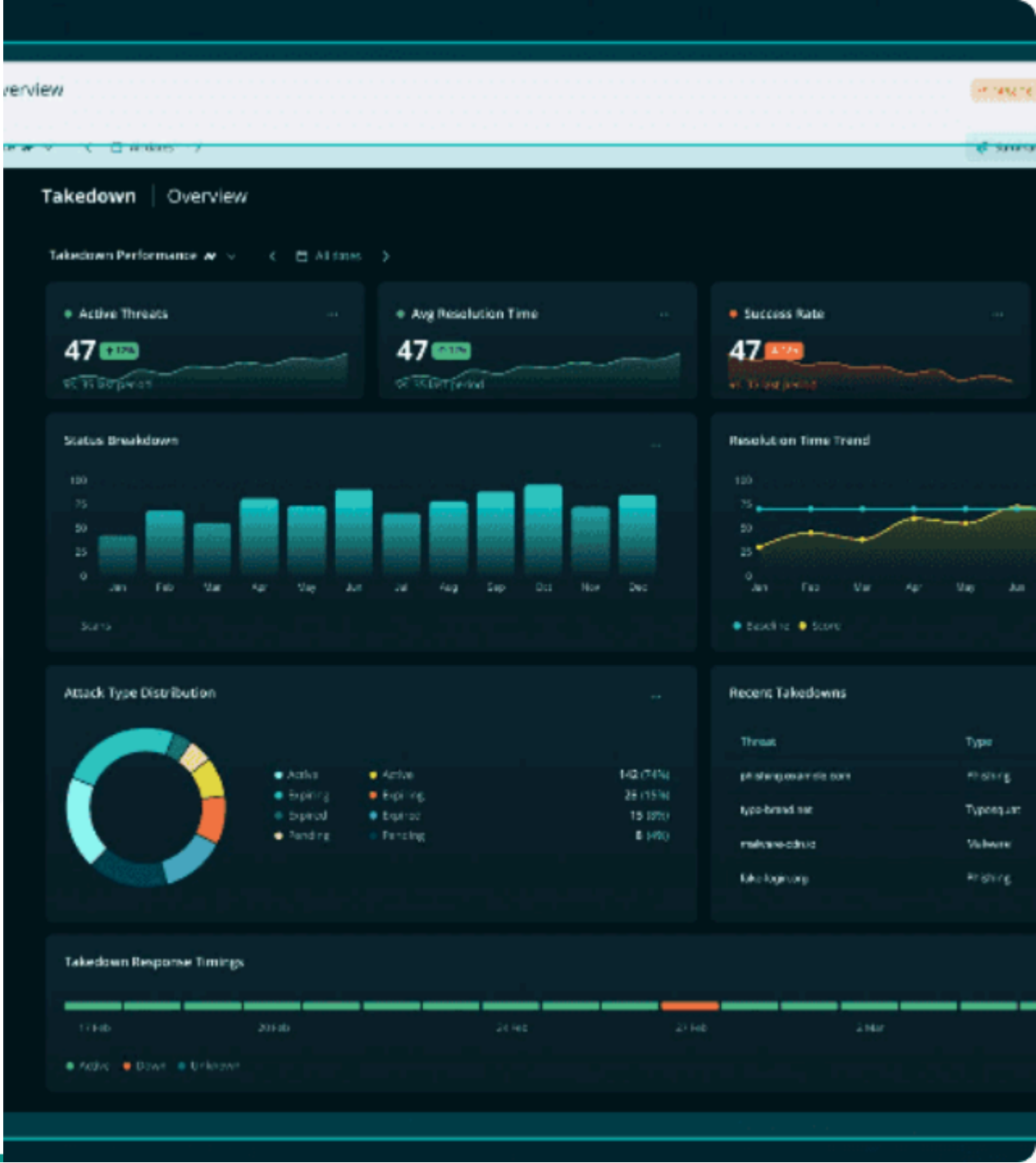
Simple & Interactive Interface

The web dashboard provides an accessible platform for users to test suspicious headlines quickly.

Example Query:

"Scientists announce breakthrough energy source..."

● Output: REAL NEWS (Confidence: 96.4%)



Advantages & Applications

⚡ Key Advantages

- **Instant Verification:** Returns classification results in milliseconds.
- **Reduces Human Effort:** Eliminates manual initial screening.
- **Scalable:** Capable of analyzing large textual datasets simultaneously.
- **User Friendly:** Requires no programming knowledge for end users.

📈 Real-World Applications

- **Social Media Moderation:** Pre-filtering suspicious viral posts.
- **News Aggregators:** Verifying story credibility before publishing.
- **Educational Portals:** Teaching digital literacy and source evaluation.
- **Browser Plugins:** Real-time background news verification.

Limitations & Future Scope

! Current Limitations

- Accuracy is dependent on dataset quality and coverage.
 - Sarcasm, satire, and subtle context are challenging for basic models.
 - AI assists but cannot entirely replace human fact-checking context.

🚀 Future Enhancements

- Integrate Deep Learning models like BERT / RoBERTa.
- Multilingual support for global news verification.
- Multimodal detection (analyzing news text along with images and videos).
- Real-time web browser extensions.

Thank You!

AI-Based Fake News Detection

System
Questions & Answers

Image Sources



https://www.crnasia.com/india/news/2026/media_1bdf6099561320e30f89e520679bd1e355613fd85.jpg?width=750&format=jpg&optimize=medium

Source: www.crnasia.com



<https://framerusercontent.com/images/jB3qjxwvFZ6u8i742uQy6LuJmcA.png?width=879&height=533>

Source: www.netcraft.com